Satellite-based flood, debris and accessibility mapping for disaster response

Evidence synthesis

The literature supports a practical pipeline—map water and observable damage from satellite data, translate mapped impacts into network disruption, then expose uncertainty and provenance to decision-makers—but it does **not** support treating these as one already-solved end-to-end task. Flood segmentation has public benchmarks and reproducible baselines, while event-independent evaluation, verified infrastructure damage labels and image-to-accessibility integration remain less established.[1][2][3] The matrix accompanying this brief curates 19 papers spanning benchmark, method, infrastructure and operational-accessibility evidence.

1. Sentinel-1 SAR flood segmentation and change detection

The most useful early shared baseline is Sen1Floods11: 4,831 georeferenced 512×512 chips spanning 11 flood events, six continents, and multiple biomes/ecoregions. Its original FCNN comparisons show that flood-specific labels matter, and that event-derived optical labels can scale training beyond scarce hand labels; backscatter thresholding remains a useful simple comparator, especially for permanent water. The dataset and training/evaluation code are public.[1]

A stronger operational comparison evaluates AlbuNet-34, FCN, DeepLabV3+, U-Net and U-Net++ against a rule-based Sentinel-1 processor. Dual-polarization inputs improved IoU by five percentage points relative to the rule-based chain, a weighted cross-entropy/Lovász combination added two points, and radiometric augmentation helped whereas geometric augmentation hurt. Two flood-event tests reached IoU 0.96 and 0.94; these results should not be compared naively with other papers' benchmark IoUs because event, label, water-class and split definitions differ.[4]

For a student baseline, compare thresholding (or an operational processor, if available), Random Forest, and a U-Net on Sentinel-1 VV/VH, and report per-event precision, recall, F1 and IoU. Dual-polarization and event-level testing are represented in existing benchmarks.[4] Include permanent-water exclusion and a pre-event reference where available; pixel-level random splits do not establish transfer.

2. Multitemporal SAR and optical fusion

Fusion has a clear physical rationale: SAR is available through cloud and darkness, while optical reflectance can help distinguish water classes and land-cover context. On Sen1Flood11, a Sentinel-1/Sentinel-2 deep-learning study used focal loss and reported test IoU 52.30%, mIoU 52.99%, and overall accuracy 92.81%; its Bolivia evaluation had IoU 76.74% and mIoU 47.88%. The spread across metrics and test subsets illustrates why class definition and split details must accompany headline scores.[5]

A temporally dense Sentinel-1/Sentinel-2/DEM case study of the 2018 Baige outburst flood in Tibet used index thresholds in cloud-free areas and SAR features plus terrain in cloudy areas, with a Random Forest prediction stage. Field/confusion-matrix validation reported 92.3% overall accuracy and Kappa 0.89, and the full event extent was produced within five days. This is valuable mountainous-event evidence, but it is a single case rather than a cross-region generalization test.[6]

GEOID-Flood, reported in 2026, is a promising newer benchmark: more than 14,000 tiles from 219 events in 65 countries, with co-registered pre/post Sentinel-1, pre-event Sentinel-2 and DEM, and manually validated labels. Its abstract reports modest but consistent foundation-model advantages, best transient-water resolution from finetuned SAR-optical fusion, and improved unseen-event transfer relative to earlier datasets. Exact split definitions and numeric metrics were not verified in the available record, so those claims should be checked against the full paper before operational reliance.[7]

3. Flood and debris in mountainous terrain

Mountain terrain raises distinct failure modes: layover/shadow, steep-slope drainage, vegetation, roughness changes, and rapidly changing debris or outburst water. The Baige case above shows a workable Sentinel-1/2/DEM collaboration strategy. Earlier debris-hazard work fused RADARSAT-1 SAR with SPOT-5 optical/3D mapping for Taiwan slope failures and compared delineation/volume estimates with field measurements, but the retrieved abstract gives no numerical accuracy.[6][8]

The retrieved SAR landslide/debris studies are individual-event demonstrations: the Baige outburst flood is one mountainous case, while the Taiwan debris-hazard study compares fusion outputs with local ground truth.[6][8] This evidence supports terrain context as a testable input, not a claim that one architecture is established as superior.

4. Transfer learning and generalization to unseen events

Randomly held-out tiles from the same event are a weak proxy for deployment. Geographic location, incidence angle, land cover, flood dynamics and label sources shift the input and target distributions. A Sen1Floods11 domain-generalization study treating flood events as separate source domains reports about a 3% IoU/F1 gain.[9]

UrbanSARFloods adds 8,879 512×512 chips from 18 events across five continents, with Sentinel-1 intensity and coherence. Its abstract emphasizes persistent urban-water difficulty and says weighted cross-entropy and pretrained transfer learning do not by themselves resolve class imbalance and limited training data.[10] GEOID-Flood's event diversity and reported unseen-event transfer are promising, but should be independently checked using explicit leave-one-event-out or leave-one-region-out splits.[7]

5. Road, bridge and building damage estimation

Damage evidence spans markedly different sensors and spatial scales. A 2024 tiered bridge study combines Sentinel-1 coherence difference, crowdsourced information and high-resolution imagery with semantic segmentation, and validates a case containing 17 bridges in Ukraine. It demonstrates a multiscale assessment design, but is not a validated satellite-only classifier for flood-damaged bridges.[2]

A post-disaster road-accessibility framework combines pre/post satellite semantic change detection with OpenStreetMap and is demonstrated on Indonesia's 2018 tsunami; retrieved abstract-level evidence does not provide comparable precision/recall or event-transfer results.[11] Work on automated road debris, flood depth and building damage from airborne LiDAR/aerial imagery demonstrates the importance of higher-resolution data, but is not direct evidence that medium-resolution Sentinel imagery can estimate those asset-level damages.[12]

Accordingly, distinguish *hazard exposure* (a road intersects a flood mask), *observed image change*, and *verified structural damage/passability*. Existing bridge work combines SAR coherence with crowdsourced and high-resolution imagery, while road access methods use OSM and road-status information; the available studies do not support treating a satellite cue alone as verified passability.[2][13]

6. Road-network disruption and settlement accessibility

The access literature connects flood extents or road-status reports to network/raster travel-time models, criticality, and service access. In Cyclone Idai, Mozambique, raster and network methods differed by roughly 300,000 people estimated to have lost healthcare access (about 800,000 versus 500,000); incomplete road mapping affected network estimates more strongly. The study also identifies critical and backup road segments.[3]

A Sentinel-1-based Mandalay study links flood extents to network redundancy and healthcare access, while a Papua New Guinea earthquake system combines OpenStreetMap and humanitarian road-status data for graph-theoretic reachability. These are complementary: one estimates hazard intersection/criticality, the other uses reported road condition.[14][13]

A short project can use OSM roads, settlement points and clinics, intersect roads with a flood mask, then compare baseline and disrupted shortest-path travel time or reachable population. This connects satellite flood maps to network criticality and access measures used in existing studies.[14][3] Make blockage assumptions explicit and test alternatives; distinguish modelled exposure from observed closure.

7. Uncertainty, false positives and decision support

SAR false alarms can arise from permanent water, radar shadow/layover, wet soil, urban double-bounce and registration error; cloud and shadow challenge optical inputs. Fusion can reduce some ambiguities but also introduces co-registration and missing-modality problems. The retrieved uncertainty-aware 2D/3D change-detection work explicitly attaches uncertainty/ambiguity to segmentation and 3D reconstruction, with expert review in earthquake and wildfire settings, but provides no flood-specific calibration evidence in its available abstract.[15]

For flood mapping, uncertainty should be tested rather than represented only by a confidence-colored map: calibrate probabilities on held-out events, inspect false positives by land cover and slope, report precision-recall trade-offs, and include an abstain/review class for low-confidence areas. Uncertainty-aware 2D/3D change detection exists, but the retrieved record does not establish a best flood-specific uncertainty method for response decisions.[15]

8. Automated, evidence-grounded situation reports

The operational evidence supports the need for interoperable, timely, confidence-aware information, rather than proving that automated narrative reports improve response outcomes. A broad systematic review of disaster situational-awareness technologies included 302 articles/after-action reports and identified delayed/unreliable communications, time-intensive analysis/visualization and weak interoperability as constraints.[16]

A defensible student prototype would generate a compact structured report from computed outputs only: event/time window, mapped flooded area, affected road length, settlements or services with changed travel times, top uncertain regions, data lineage and links to the map/evidence. This addresses documented response challenges in delayed analysis, visualization and interoperability, while avoiding a claim that generated prose itself has been shown to improve decisions.[16]

Cross-study comparison and interpretation

Reported headline scores vary from IoU around 0.52 in one multimodal Sen1Flood11 setup to 0.96/0.94 in event tests from another study; these are not a leaderboard because they differ in class targets, reference labels, geography, polarization, event selection and thresholding. Overall accuracy can hide rare flooded-pixel errors, so include classwise precision, recall, F1, IoU, confusion matrices and event-level scores.[5][4]

Dimension	Evidence-weighted reading
Inputs/preprocessing	Sentinel-1 amplitude and dual polarization are common; multitemporal intensity/coherence can improve change cues. SAR radiometric augmentation helped in one benchmark; geometric augmentation hurt. Optical and DEM inputs can help but co-registration and cloud gaps matter.[4][6]
Architecture/training	U-Net/FCN/FPN families are reproducible baselines. In one comparison architecture differences were not significant; data representation, labels, loss and split design may matter as much or more.[4]
Resolution and timing	Sentinel-1 GRD supports broad-area flood extent but not detailed structural diagnosis; finer-resolution optical/airborne data are used for asset-scale debris and damage. Flood-response timeliness is event-dependent; the Baige map was produced within five days.[6][12]
Geography and dataset size	Sen1Floods11, UrbanSARFloods and GEOID-Flood provide increasing event/geographic diversity, yet class imbalance, urban performance and label harmonization remain.[1][10][7]
Evaluation design	Event- or region-held-out testing is essential. Claims of generalization need independent events and reported per-event metrics; random tile splits alone are insufficient for deployment claims. The retrieved records often did not expose enough split detail to verify this.[9][7]

Dimension	Evidence-weighted reading
Compute/reproducibility	Public data and code are reported for Sen1Floods11, UrbanSARFloods and GEOID-Flood. GPU/runtime requirements were generally not available in the retrieved records; U-Net-level baselines are preferable for a short project over reproducing large transformer systems.[1][10][7]

Evidence-supported gaps (not novelty claims)

1. **Event-independent validation remains a central need.** Domain shift is explicitly documented; recent benchmarks are designed to improve event diversity, but many abstract-level records do not reveal rigorous leave-one-event-out performance.[9][7]
2. **Urban flooding and class imbalance remain difficult.** UrbanSARFloods explicitly reports this limitation and questions whether common weighting/pretraining approaches suffice.[10]
3. **Mountain debris/outburst floods are comparatively case-study driven in the retrieved evidence.** Multisensor terrain-aware examples exist, but available records do not establish broad transfer across terrain, sensors or event types.[6][8]
4. **Image-derived hazard, asset damage and actual passability are not interchangeable labels.** Infrastructure studies use mixed high-resolution imagery, coherence and crowdsourced inputs; network access results are sensitive to road-map completeness.[2][3]
5. **Uncertainty-to-action validation and evidence-grounded automated reporting are underdocumented in the retrieved flood-specific studies.** Uncertainty methods exist for broader disaster change detection and disaster information systems, but the link from calibrated model uncertainty to response decisions was not established here.[15][16]

Recommendation for a 15-day student project

Choose one event and one tractable output: **Sentinel-1 flood extent → road exposure → settlement/service accessibility change**, using a public benchmark event or a well-documented event with pre/post SAR and reference labels. Sen1Floods11 and UrbanSARFloods provide public starting points.[1][10] Use a simple threshold/rule-based comparator and U-Net (or RF if compute/data preparation is constrained); compare single-date with pre/post change features and optionally add DEM slope. Hold out a whole event or spatial block, not random adjacent chips. Report precision, recall, F1, IoU, confusion matrix, and false-positive examples by land cover; then convert the mask to road intersections and compare at least two closure assumptions in a graph-based access analysis. Existing access studies demonstrate sensitivity to methods and road-data completeness.[3] Keep the narrative report templated and generated from measured outputs, with a confidence/limitations section rather than open-ended image interpretation.

A realistic 15-day scope is **one flood map, one honest held-out evaluation, and one transparent access scenario**, not a new multimodal foundation model or validated bridge-damage classifier. The highest-value contribution is likely careful split design, error analysis and traceable impact translation, not claiming a new architecture. Open-code availability for Sen1Floods11 and UrbanSARFloods makes those stronger starting points than unverified recent methods.[1][10]

Review coverage and verification limits

This is an in-session evidence synthesis, not a formal registered or platform-run systematic review. Discovery used academic search queries organized around the requested topics; the delivered matrix contains 19 selected source-linked records. Most study details were available as abstracts or indexed summaries; no comprehensive full-text retrieval and extraction was completed. Consequently, papers whose full-text or split/compute details could not be verified are explicitly marked in the matrix. Particularly thin areas in this search were flood-specific uncertainty calibration, automated evidence-grounded report evaluation, and satellite-only road/building damage labels. The matrix links to DOI, publisher or available source pages where retrieved; a linked record does not imply full-text verification.

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